

Red meat consumption and cancer: reasons to suspect involvement of bovine infectious factors in colorectal cancer.



An increased risk for colorectal cancer has been consistently reported for long-time consumption of cooked and processed red meat. This has frequently been attributed to chemical carcinogens arising during the cooking process of meat. Long-time fish or poultry consumption apparently does not increase the risk, although similar or higher concentrations of chemical carcinogens were recorded in their preparation for consumption. The geographic epidemiology of colorectal cancer seems to correspond to regions with a high rate of beef consumption. Countries with a virtual absence of beef in the diet (India) or where preferably lamb or goat meat is consumed (several Arabic countries) reveal low rates of colorectal cancer. In China, pork consumption has a long tradition, with an intermediate colorectal cancer rate. In Japan and Korea, large scale beef and pork imports started after World War II or after the Korean War. A steep rise in colorectal cancer incidence was noted after 1970 in Japan and 1990 in Korea. The consumption of undercooked beef (e.g., shabu-shabu, Korean yukhoe and Japanese yukke) became very popular in both countries. The available data are compatible with the interpretation that a specific beef factor, suspected to be one or more thermoresistant potentially oncogenic bovine viruses (e.g., polyoma-, papilloma- or possibly single-stranded DNA viruses) may contaminate beef preparations and lead to latent infections in the colorectal tract. Preceding, concomitant or subsequent exposure to chemical carcinogens arising during cooking procedures should result in increased risk for colorectal cancer synergistic with these infections. Copyright © 2011 UICC.